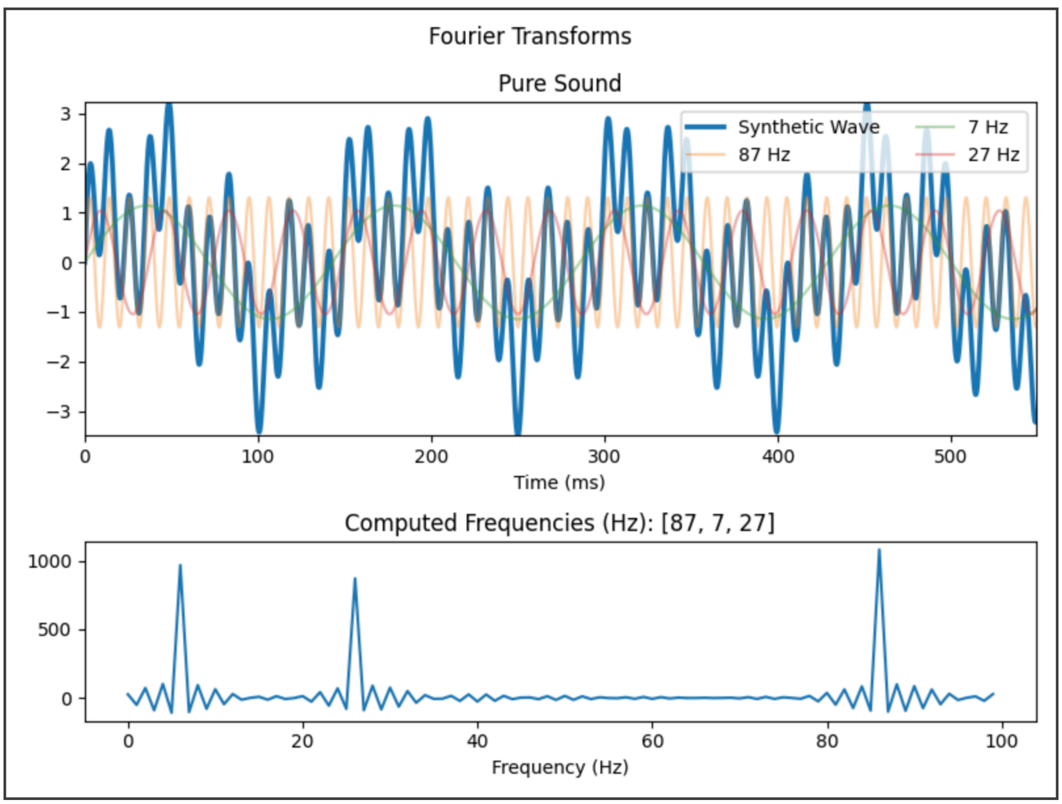
Fourier Analysis and Quantum Uncertainty

Fundamental Trade-offs in Measurement

The uncertainty curve connecting precise measurements of location and those of change of location (several kinds) is a tantalizing constraint on our ability to understand the world through a reductionist lens.

The phenomenon was initially understood in the context of the Fourier Transform: a mathematical operation to take a set of measurements from the *time domain* into the *frequency domain*. For example, given amplitudes (many per second) of the air pressure against the filament of a microphone, you can measure which frequencies (and exactly how much of each) compose the sound.



In this example, we see a signal in blue in the top chart—unknown frequency—and the pure frequencies which make it up (plotted for visual aid). The bottom is the product of the Fourier Transform (called a spectrogram), which only took in that raw signal. It deduced, correctly, that 7, 27, and 87 Hertz were the primary frequencies that composed the sound.

The transform gives very valuable information, but it loses some in the process. Whatever interval of time you transform will be reduced to an instant, in which frequencies are measured for strength, not time or location. That is, if our signal started with 7 Hz as dominant, then transitioned to 27 and 87, our spectrogram would be ignorant to it. This means that if you want to pin down *when* a frequency was present, you need to shrink your time window of observation. However when you shrink your window of observation, the *certainty* with which you detect frequencies diminishes. Let's unpack the mathematics of the Fourier Transform to get a better feel for this trade-off.

Fourier Transform Definition

While periodic sound signals (and others we'll soon discuss) have both amplitude and phase (denoting *when* periods occur, not just how strong they are), we will stick with just amplitude. This lets us skip using complex numbers. The intuition behind the Fourier Transform still survives robustly in this idealized example.

In linear algebra, there is an operation called the Dot Product (or Inner Product) and it measures what is known as the *cosine similarity* between two vectors. Vectors define specific directions in linear space and the cosine similarity measures the similarity of the directions, in terms of the angle between them. The dot product between vectors **A** and B—each of dimension N—is defined as:

$$Dot(A, B) = \Sigma_i^N a_i b_i$$

Simply, it is a weighted sum of the terms (denoting *x*, *y*, *z*, ... dimensions of space in which these vectors reside) of the vectors. Even more coarsely than computing exact angles between vectors, the dot product gives a quick reading of the alignment of the vectors. When two values of the same sign (+ or -) are multiplied, the sign stays the same. Thus terms a_i and b_i who have the same sign (are pointing in the same general direction in space) will *add* to the overall sum taken in the operation. Those dimensions where **A** and **B** disagree will reduce the overall sum. Vectors whose dot products are negative are pointing is opposing directions, those with dot products of 0 are perfectly orthogonal, and those with positive dot products are pointing in the same general direction. The greater the magnitude either way, the stronger the misalignment or alignment.

While this operation is intended for vectors pointing in space, its results can just as well explain the alignment of any two time series (which are just lists of values, like vectors!). Thus we can measure the degree of *harmony* or *accord* between any two time series of the same length with the dot product.

The Fourier Transform, then, is the natural application of the dot product to the domain of matching periodic series with the pure frequencies that make them up.

If we want to take a signal, S, and find which frequencies from 0 → 100 Hz make it up, we can take the following Fourier Transform:

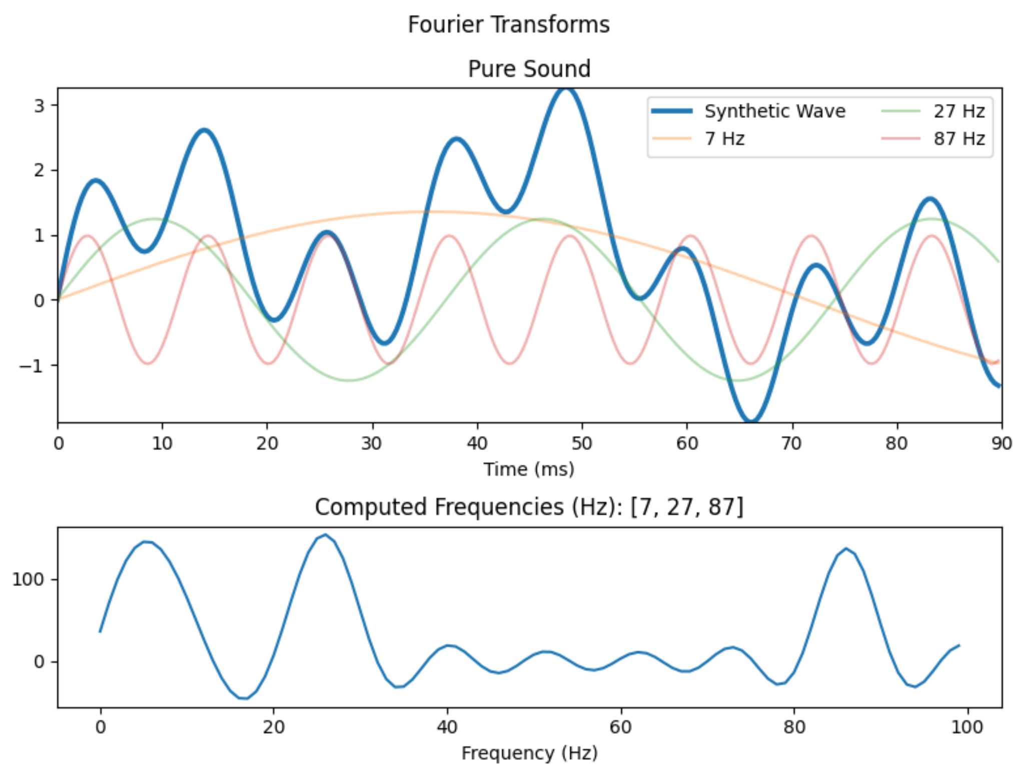
$$Fourier(S) = F \cdot S^T$$

Where F is a 100-row matrix whose rows are pure cosine waves of frequencies, f, from 0 → 100, each defined: $cos(2\pi \times f)$. The output will be a row vector with 100 terms, each representing the dot product between S and each pure frequency! The second plot in the diagram is exactly this.

Fundamental Trade-Offs in Measurement

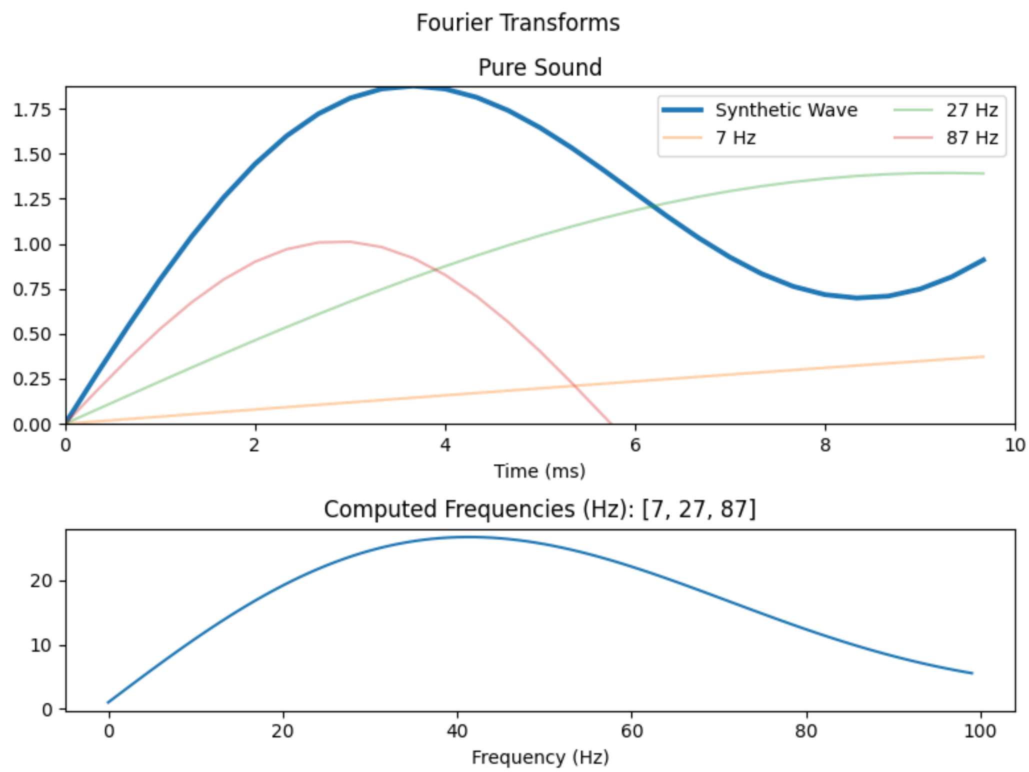
If we think critically about the operation done to produce the spectrum, we can reason that the longer the signal, S, the more terms will compose its dot product with each frequency, f_i. For one of the frequencies present in our example, say 27, every single time S and f_{27} ebbed and flowed together increased their sum. The value of their dot product is directly proportional to the number of periods in S.

If you look carefully, you'll see that even in our first diagram, the sample was only approximately 0.5 seconds. If we take the exact same signal and only sample 0.1 seconds (a more precise time and place), we get a far less precise spectrum, with dot product *around* the real frequencies being high as well:



Same exact signal, much more precise notion of *when* in the signal we're measuring, but necessarily weaker peaks across the board in the spectrum. Looking at the green signal—27 Hz—we see how it harmonizes with our signal at some points. This is what still nudges the 27-th value in the spectrum high. But you could imagine the frequencies just higher and lower than 27 (pause and literally picture what these curves would look like next to the green one…just barely looser and tighter, respectively) similarly rhyming with S. The peak is still at 27, but it is far less sharp than in the former diagram.

If we keep getting a more precise slice of time (say to 0.01 seconds), the effect is further accentuated:



Our reading on frequency evaporates; the spectrum flattens.

Intuitive Break

There are two key dynamics causing this fundamental trade-off in precision. The first is inherent to the Fourier Transform operation itself: when we sum across points that originated at specific instants in time, we *lose their temporal information*. Summing is a forgetful operator in this sense, because of its commutativity. That is, the *order in which* terms are summed together naturally does not matter! The only way to pin down location for the Fourier Transform is to shrink the measurement window, because within a window, time is forgotten.

The second dynamic is much more fundamental to the measurement of *change* in a system. Change is inherently—definitionally—a measurement across time. To see how something changes across time, you must observe it at *multiple instants*.

Taking the derivative at a point—the *instantaneous* rate of change—is only possible when you know the full definition of the function over its domain. When processing an unknown signal, we do not have such information; we have to derive it by taking discrete measurements.

This fundamental limit in measuring *both* time/location and *how* change happens (could be momentum, slope, frequency) is especially exposed when looking for frequencies of periodic functions. We don't want to just know how the line changes each instant, we want to know *how long it takes for a full cycle* to complete in the line. This simply demands more measurements.

There are philosophical insights we could pause here to ponder. The limits of our ability to formalize and measure (broadly, to *know*) are tantalizing and have defined some of the most important results in many fields of science and math. Gödel and Turing discovered the limits of what could be computed by finite algorithms. This gave birth to computing.

Before this, Werner Heisenberg discovered that any gains we make in identifying the *location* of quantum particles is financed by a less precise identification of its *momentum*. This key moment defined and sparked a century of revolutionary atomic-physical breakthroughs. He wrestled with the same trade-off that limits Fourier Transforms.

Quantum Uncertainty

The mechanics of subatomic—quantum—particles are very mysterious and unintuitive. Whereas the history of physics can be seen as one of *zooming in* and *zooming out,* defining the objects that make up our world and those we compose. The *zooming in* is reductionism: defining the world in terms of ever smaller parts. This program has continued to vindicate the intuitions of many who believe that the world is compositional. But when we hit the subatomic level, our intuitions became impotent.

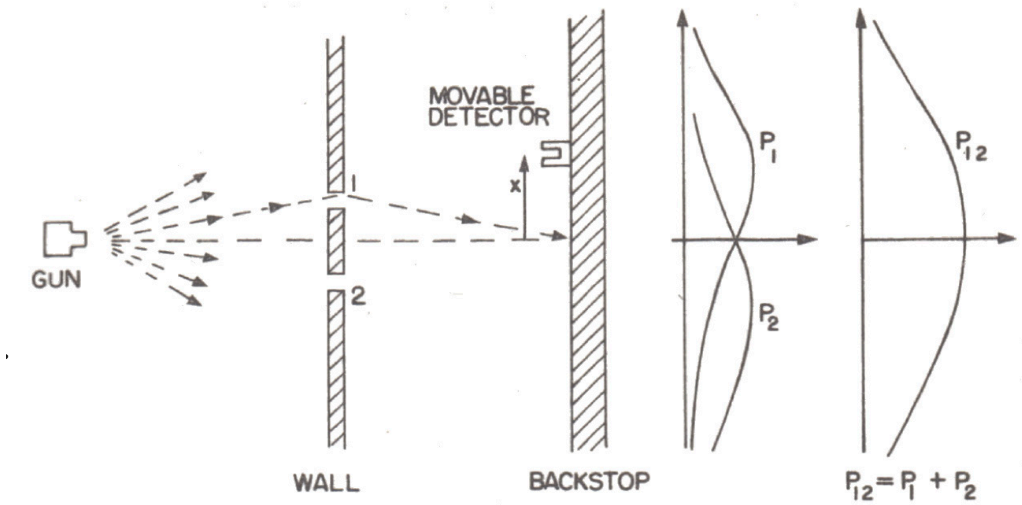
Several experiments—theoretical and physical—showed that, contrary to every other kind of physical system we've ever known, quantum particles behave at times like discrete objects and at other times like *waves*.

A principally famous experiment is known as the Double Slit Experiment. It imagines a stream of electrons, photons (packets of light), or any other quantum particle flying directly towards a wall. Between the source of this stream and this wall is another wall with two slits towards the middle. The particle flies towards these slits and either goes through the left slit, the right slit, or does not make it through.

To measure what makes it through, we can put a kind of particle detector at any arbitrary spot on the back wall. To measure through *which slit* a particle moves, we could also attach detectors in each of the slits.

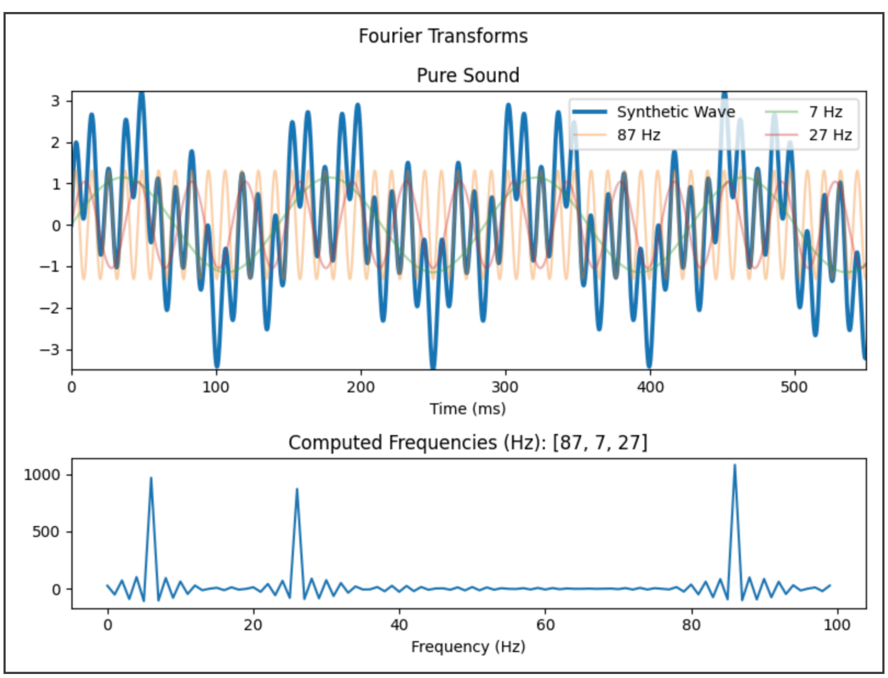
Measuring just where one particle lands on the back wall or through which slit it moves is not what we're interested in. We'd like to know *statistics* about how this system behaves with many examples. So we set up the experiment. The results were baffling.

First we run the system measuring everything. We see a particle go through the left slit, we see it hit the wall just left of the middle. We see a particle go through the right slit, we see it hit the wall just right of the middle. After many rounds, we see that the particles go through each slit an equal amount of times, and the detector on the back wall showed just what we'd expect: two normal distributions centered just left and just right of the middle of the back wall, indicating the two regions where the particles were likely to land. When they're added together (accounting for both possibilities) we get a single, flat distribution around the center of the back wall.



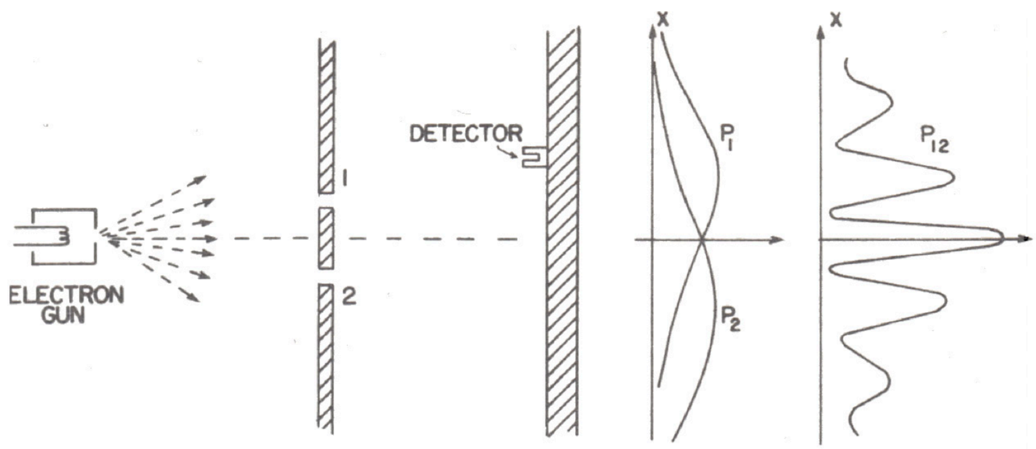
The experiment was then run ignoring through which slit the particles traveled. Thousands of particles are flung towards the wall, the detector was moved around, measuring where each particle landed. When the net distribution of points of contact on the back wall were constructed, though, the bimodal distribution from the first experiment vanished. In its place was a distribution perfectly representing a *wave interference pattern*.

Let's get some wave-interference inspiration from our first Fourier Transform. Look at our signal, *S*, from the original Fourier Transform we took:



When we have multiple frequencies interfering with each other, the resulting signal is not necessarily obvious. Sometimes 7 and 27 Hz are both positive or negative, *constructively interfering* to boost or retract the net signal, and other times they're in disagreement, *destructively interfering* to cancel the signal towards 0.

What was seen on the back wall of the double slit experiment was a pattern that looked something like *S*. It did not uniformly rise and fall—as the two bell curves did when all slits were monitored—but instead it showed an interference pattern. Reverse engineering the pattern, it ended up *perfectly fitting* the imagined scenario that instead of particles streaming from the source, through the slits, and to the wall, a *wave* was emitted from the source, with complementary parts of its wake going through both slits, being warped, and then interfering with each other on their way to the back wall.



This miraculous finding indicated that *when their position through the slit is not measured,* the particles behave like waves. You'll notice the distributions P_1 and P_2 look the same in each version of the experiment.

However, whereas in the first version of the experiment P_1 and P_2 are summed like distributions (standard practice for computing a net probability of two possibilities) and justify the final experimental results of P_{12}, in the second version, P_1 and P_2 do not justify these results in the same way.

When we *square* the apparent sum, we get $P_1^2 + P_2^2 + 2P_1P_2$. The first squared terms compose to the P_{12} ***probability distribution*** from the first experiment. This suggests that the amplitudes of these theoretical waves *encode* probability (particularly its square root). Strictly, P_1 and P_2 are actually defined as follows:

$$P_1 = \phi_1^2 , \quad P_2 = \phi_2^2$$

And our expanded eventual value P_{12} is defined as:

$$P_{12} = \phi_1^2 + \phi_2^2 + 2\phi_1\phi_2$$

These squared terms still do not tell the whole story. They bring us to the original flat distribution that is *not* observed in the actual second experiment.

This final term—though I am hiding some of the details so as to stay above water in this demonstration—represents information about the *phase* of these wavelike patterns. Further the *phase* of this theoretical wave directly corresponds to the *momentum* of the actual particle. This $2\phi_1\phi_2$ term provides for the interference regions we see where P_{12} fall towards 0, indicating phase of the theoretical wave, indicating momentum of the actual particle. It makes sense that it involves phase and interference, because it looks like the terms in a dot product! When ϕ_1 and ϕ_2 agree, their product is *constructive*, whereas when they disagree, their product is *destructive*.[1]

[1] You may then ask where *phase* was in our original Fourier Transforms? In those, we said that the differences in *frequency* caused the complicated interference pattern in S. But now we're saying that the *phase* of ϕ_1 and ϕ_2 causes the interference. In truth, the composition of two waves and their interference is determined by both *frequency* and *phase*. These two examples were deceptively complementary because in the former, we implicitly had *phase* fixed and the latter had the *frequency* fixed. In our Transform, all frequency rows, f_i, in our matrix started at 0. There was no difference in phase between any f_i and S. In the latter, alternatively, the *frequency* of the two waves coming out of the slits were fixed. The only difference in the two waves that propagated from each slit was their spatial orientation. Two signals of the same frequency can cancel each other perfectly when their phase difference is π, or half of a period. Frequency-induced patterns have to do with the changes in rates of oscillation of the waves—literally how often they go up and down. Phase-induced patterns, for any pair of frequencies, have to do with *where* peaks and troughs occur.

We will not dwell too much on the mathematics. From here, though, we obtain the dumbfounding insight that particles behave like waves when they're not observed. In other words, when we know their position with less precision, we have a better understanding of their momentum. If we hone in on tracking the position of the particle, we lose the interference pattern and the implicit information about the particle's momentum.

Quantum → Waves → Fourier Uncertainty Returns

While we are not trading *frequency* information of a signal for *time* information of that signal, we are trading *momentum* information for *location* information. There are parallels, though. Pinning down the *location* of something depends on time being frozen. Pinning down the *momentum* of something depends on looking across time, just like for *frequency*.

The parallels are made more explicit by the fact that the way we actually define the state (all measurements, *location, momentum,* and others) of quantum particles is with periodic wave functions. *Momentum* is literally encoded as a *frequency* in this probabilistic wave function. *Location* is derived from letting this function unfold over time.

In both pure signal processing and quantum measurement, we have waves as the fundamental representation. Thus we have the fundamental Fourier-informed uncertainty trade-off between location in time and frequency across time. It is as if when we seek to *pause* the system to get a precise reading at an instant, we collapse the *cross-temporal* information defining that system.

The uncertainty of quantum systems is not because they're so small (or some other reason), it is because they exhibit wavelike properties. Treating the system as a periodic one defined by waves is the only way to mathematically justify our experimental results. Embracing the waviness of matter (why ever it is the proper mathematical formalism for matter), we come up against Fourier's fundamental uncertainties.

Thus the most reduced form of matter exists in a slippery environment with fundamental—clear, intuitive—limits of measurement.

***More Slits*: Feynman's Takes Us Further**

Learning about this double slit phenomenon, Richard Feynman was eager to push it further. Seeing that the particles, instead of just choosing one path and taking it, avoided paths where their waves would interfere, he wondered how these particles could *know* about this interference? It seems they had information that traditional causality would never permit them to have.

Did they simultaneously take both paths, see where they interfered, then chose a route that made it to the wall?

What if we added a third slit? Just as with two slits, we have the original wave function from the source, ϕ, splitting into its three warped waves, ϕ_1, ϕ_2, and ϕ_3, then composing a complicated interference pattern, P_{123}. Instead of taking into account *two* waves/paths, P_{123} reflected the interference of *three*. Why stop there?

As we take the number of slits to ∞—in the context of the experiment, meaning effectively *no* slits—what happens?

Well, as your normal intuitions would then suspect, the particle simply follows a straight line. The distribution $P_{1\ldots\infty}$ was simply a vertical line with the height of 1 in the center of the wall. Are there infinite slits and infinite ϕs interfering to leave only this simple path? Does the particle *explore* these infinite paths and choose (what seems like) the simplest one? Or is this notion of exploration just a fantasy?

Well the 1, 2, and 3 cases make the interference, path-exploration notion necessary! So it really seems that when there are infinite/no slits, the particle explores an infinite set of paths which all interfere in such a way as to only magnify the path with the shortest distance.[2]

This phenomenon has been confirmed in many domains, far more specific and intricate than the slit experiment.

Conclusion

Wave representations introduce fundamental limits to measurement precision. Not because of our practical tools, but because of mathematical necessity. Quantum particles—almost in a bid to tease LaPlace's Demon—are only understandable in general when we represent them as waves, relegating their precise measurement to a world of uncertainty.

To me this is somewhat humbling and certainly very mystical. Just as Gödel and Turing discovered limits to formal understanding, our best way to model the smallest particles we know of introduces limits to our *physical* understanding.

But it stems from a similar limit in our ability to reason.

Everything we intuitively understand about the physical world is predicated on the notion of discrete objects in space and a forward march through time. We find that at base, these two elements—location and time—are exclusionary of each other. They can only be understood as a singular fabric, impossible to be pulled into separate threads.

---

[2] It is not really *distance*, but *Action*. $A = \int (KE - PE)\, dt$, where *KE* is kinetic energy, *PE* is potential energy, and *dt* is the change in time. When there is no potential energy acting on the particle, this reduces to $A = \int m\, v\, ds$, where *m* is mass, *v* is velocity, and *ds* represents infinitesimal portions of a path. Integrated across these small portions of the path, this becomes something like *work*, something like *distance weighted by mass and speed*. For the purposes of our idealized ∞ slit experiment, with free particles, the only term that matters is *distance.*